MIGRATION HAND-OFF

Addressbook, Vaadin 8 to Vaadin 25

Hand-off document

FINAL, ON THE CUSTOMER'S ANSWERS OF 23 OCTOBER 2026 (SECTION 7)

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Prepared for	Product owner, operators and developers of the Addressbook app

Every decision in section 7 was made by the customer on 23 October 2026; no decision is open. This document is generated from its Markdown source (docs/handoff.md).

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1 Executive summary

Outcome. The Addressbook app now runs on Vaadin 25 (25.3.3, the free `vaadin-core` components only), Java 21 and Jetty 12.1. It has the same single screen as before: a filter field, a "New contact" button, the contact list, and a form with Save and Cancel that opens when a contact is selected. The data is still the in-memory demo list of 100 contacts. All 11 user workflows recorded on the old app were run on the new app step by step: 11 of 11 meet the agreed acceptance bar. Eight give the same results step for step; the other three each show a small difference, which you accepted on 23 October 2026 (section 7). The app is built, started, smoke-checked and tested end to end in CI on every push.

What changed.

- For users: a new look (Vaadin's Lumo theme instead of Valo) with the same layout, fields, buttons and data; a new calendar popup; small differences in messages and keyboard behaviour (section 2.1).
- For operators: a Jakarta EE 11 / Servlet 6.1 container (Jetty 12.1) on Java 21 instead of Jetty 9.2 on Java 8 bytecode; production mode is now decided when the war is built; a new set of runtime dependencies under Apache 2.0 or MIT; JUnit, used only by the tests and not shipped, is EPL 2.0 (section 2.2).
- For developers: Vaadin Flow's Java API instead of the Vaadin 8 API and its Vaadin 7 compatibility components; 7 unit tests and 10 end-to-end tests, where the old app had none (section 2.3).
- You accepted all of these changes; sections 2.1 to 2.3 say when each was decided, and section 7 lists your decisions of 23 October 2026.

Risks.

- commons-beanutils 1.9.2 stays, as agreed, with two known vulnerabilities (and one more in commons-collections 3.2.1, which it pulls in). The app only copies its own contact objects with it, so exposure is low. Removing it is a small follow-up (section 6).
- In production mode a server-side error is not visible to the user or in the browser console; only the server log shows it. CI fails on any error in the server log.
- The end-to-end tests wait on an internal Vaadin client object; a future Vaadin update can make them flaky without warning (section 6).
- The birth-date workflow has no end-to-end test yet. It was checked step by step against the old app, but CI does not run it. Adding that test is the first follow-up (section 6).

Decisions made. On 23 October 2026 you answered all 21 items, each as recommended (section 7). On the calendar, the difference is accepted and the prepared fix is not taken. The eight small behaviour and look changes, including the missing browser tab icon, and the eleven changes to how the app is built, run, logged and tested, what it is built from, and its layout, are accepted; usage statistics stay off. The migrated app stays on `mig-dev`, beside the old app on `main`. No decision is open.

2 What changed

This chapter lists only the changes recorded in the migration's change register. Each change says when you accepted it: at an earlier milestone, or with your answers of 23 October 2026 (section 7).

The migration started from your app on Vaadin 8.0.0.alpha2, a pre-release that is only available from a Vaadin pre-release repository. The first step was to bring it to Vaadin 8.14.5 (a free Vaadin 8 release from Maven Central) without other changes, and to add CI and a Docker environment. That stabilised Vaadin 8 app is the reference the new app was compared with. It is on the `main` branch of the output repository; the migrated app is on `mig-dev`, and stays there (your decision of 23 October 2026).

2.1 For users

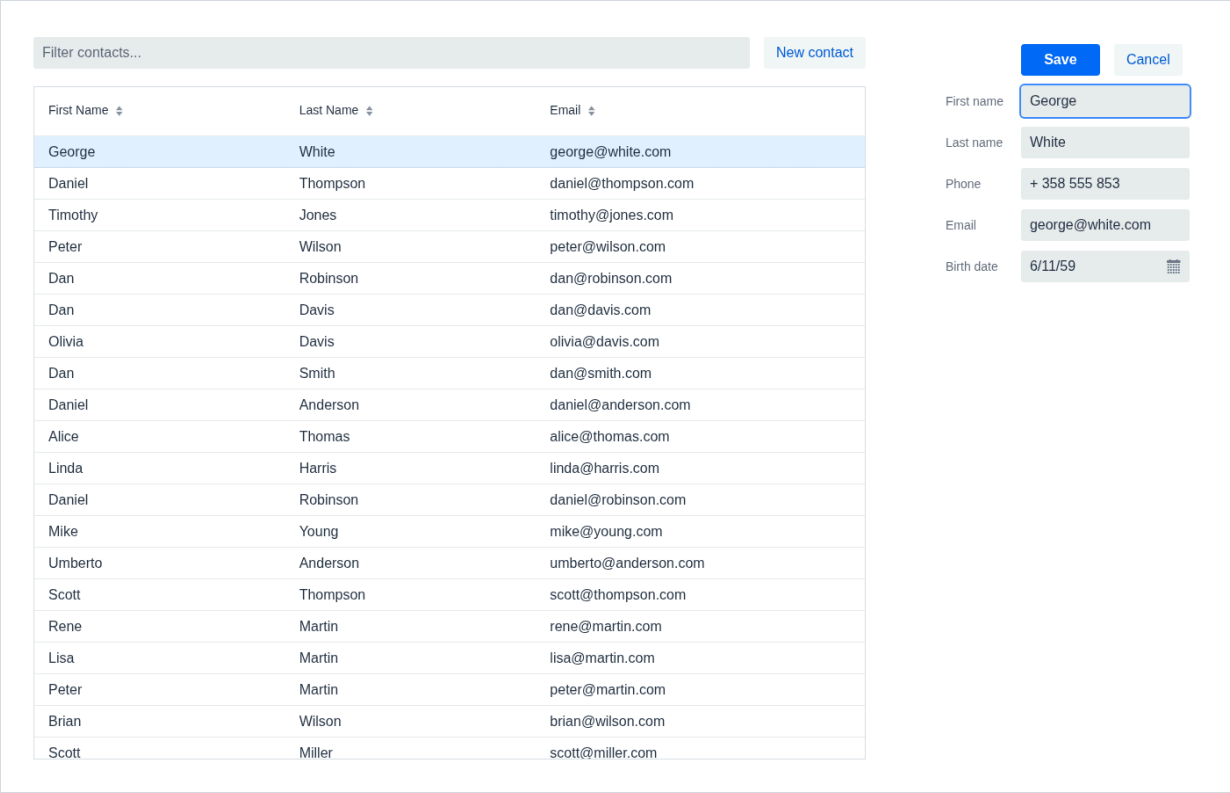


Figure 1. The new app with a contact open (Vaadin 25, Lumo theme), 1400 x 900 window.

Area	Old (Vaadin 8)	New (Vaadin 25)	Status
Look	Valo theme	Lumo theme: grey-filled fields, plain buttons, no grid column lines, pale row selection, system font, white background. Same layout: list on the left, a one-column form with labels to the left of the fields on the right, Save and Cancel above the fields, the same margins. About 21 rows visible at 1400 x 900 (old 19.5)	Theme chosen at the second milestone (Lumo). The restored layout: accepted on 23 October 2026
Birth date in the alpha	Dates shown with a trailing comma, typed dates rejected	Fixed by the Vaadin 8.14.5 upgrade and kept: `6/11/59`, typed dates save	Old bug not copied

Area	Old (Vaadin 8)	New (Vaadin 25)	Status
Birth date field	`M/d/yy`, typed dates accepted	Same format and typing. A typed two-digit year in the boundary year is read by year, not by day (example: `10/22/46` typed in 2026 is stored as 1946, the old app stored 2046). The date order is always the US one, also for an en-GB browser	Accepted on 23 October 2026
Birth date after Save	Saving an untouched contact kept the time of day of the demo data	Save stores the birth date at midnight, so text that matched a time of day in the filter no longer matches a saved contact	Accepted at the first milestone
Invalid birth date	Red border and "!" marker; the message only as a tooltip	"Date format not recognized" under the field as soon as you leave it. Save does nothing in both	Accepted on 23 October 2026
Calendar popup	One month with arrows	Scrolling months with a year column, Today and Cancel buttons. Same function	Accepted on 23 October 2026
Calendar for a new contact	Opens on the current month	Opens on the month last shown in this browser tab (for example June 1959 after looking at George White)	Accepted on 23 October 2026; the prepared fix is not taken
Enter with the calendar open	First Enter picks the day, a second Enter saves	One Enter picks the day and saves	Accepted on 23 October 2026
Clicking a field's label	Did nothing	Puts the cursor in the field	Accepted on 23 October 2026
"Saved" and "Cancelled" messages	Stayed until the mouse moved or a key was pressed, then closed about 3 s later	Close by themselves after 3 s. Same text and place (bottom right)	Accepted on 23 October 2026
After a server restart or outage	"Session Expired" or "Communication problem" messages	No message after a restart; the next click reloads the page and is lost. During an outage a striped bar "Connection lost, trying to reconnect..." shows, and the page reloads by itself when the server is back	Accepted at the second milestone
Other addresses	Any path showed the screen	Still any path shows the screen; a trailing slash is dropped from the address bar (`/foo/` shows as `/foo`). `/VAADIN/` answers 404 (old 403); `/HEARTBEAT/` shows the screen (old 403)	Accepted at the second milestone
Browser tab icon	Vaadin's default icon from the Valo theme	The browser's default icon (the app never had its own)	Accepted on 23 October 2026; adding an icon is follow-up work (section 6)

2.2 For operators

You accepted every change in this table on 23 October 2026 (section 7.2), except the parts a row marks as accepted earlier.

Area	Old	New
Framework	Vaadin 8.0.0.alpha2 (customer's original), then 8.14.5	`vaadin-core` 25.3.3 (Flow 25.3.4), Apache 2.0. No commercial components and no licence key needed

Area	Old	New
Servlet container	Jetty 9.2.3 (`jetty-maven-plugin`), `javax.servlet` 3.0	Jetty 12.1.14 `ee11` (`jetty-ee11-maven-plugin`), `jakarta.servlet` 6.1. Deploy on a Jakarta EE 11 / Servlet 6.1 container
Java	Java 8 bytecode	Java 21 (`release` 21); build and run on JDK 21
Production mode	Hard-coded off in the code; only the env and CI forced it on at start	Decided at build time: the war from `mvn package` always runs in production mode; `mvn jetty:run` is development mode. The env and CI switch (`APP_ENV_PRODUCTION_MODE`) keeps its meaning but now rejects any value other than `true` or `false`
Dependencies	jsoup 1.14.3 (a known vulnerability), the unmaintained Vaadin 8 line with its warning banner, the `vaadin-prereleases` and `vaadin-addons` repositories	New runtime jars under Apache 2.0 or MIT: Jackson 3, Atmosphere, Reactor, collaboration-engine, Lumo and Aura; jsoup 1.23.2. The servlet API, which the server provides, is EPL 2.0 or GPLv2 with the Classpath Exception. `jspecify` in the war moves from 1.0.0 to 1.0.1 (annotations only, Apache 2.0), because the test library needs the newer version. Test only, not in the war: JUnit 6 (EPL 2.0) and Playwright for Java (Apache 2.0). Only Maven Central is used. commons-beanutils 1.9.2 unchanged
Logging	Jetty's and Vaadin 8's log lines	`slf4j-simple` 2.0.17: Vaadin's lines as `[thread] LEVEL logger - message` on stderr. Every start logs three `[WARNING] The XML schema [...] could not be found` lines from Jetty 12; they are harmless and the log check ignores them
Docker environment	Image `addressbook-old` , start about 2.5 s	Image `addressbook-new:<commit>` (about 1.4 GB), same `env/up.sh` , `reset.sh` , `down.sh` ; every production-mode start and reset rebuilds the war offline and runs the unit tests first (about 5 s); a failing test stops that copy from starting (accepted at the second milestone). Development-mode copies (`APP_ENV_PRODUCTION_MODE=false`) do not run the tests
Dev-mode tools	`?debug` window	Vaadin's dev tools and Copilot in development mode only; none of it is in the war. Usage statistics are switched off for every development run (kept off, decided on 23 October 2026). Copilot checks its release notes at copilot.vaadin.com (only the Vaadin version is sent; confirmed at the second milestone); when offline this logs an error in development mode

2.3 For developers

You accepted the changes below on 23 October 2026 (section 7.2), except the routes, accepted earlier.

- **API:** every class uses Vaadin Flow (`com.vaadin.flow.*`) instead of the Vaadin 8 API (`com.vaadin.ui` : layouts, buttons, notifications) and its Vaadin 7 compatibility components (`com.vaadin.v7` : grid, text and date fields, field group): `Grid` , `FormLayout` , `Binder` , `DatePicker` , `Notification` . The four app classes keep their names and roles; `AppShell` is new and loads the Lumo theme.
- **Routes:** the screen is at `/` and at any other path (`@RouteAlias(":path*")`), as the old servlet mapping did (accepted at the second milestone).
- **Build:** `vaadin-maven-plugin` 25.3.3 adds a frontend step to every `mvn package` (it uses Vaadin's pre-built production bundle, so no Node run is needed). `maven-compiler-plugin` 3.14.1, surefire and failsafe 3.6.0 pinned. `vaadin-dev` is an optional dependency, so it is on the development classpath only.
- **Tests (new):** 7 JUnit 6 unit tests of the backend (storage returns copies; the 100 demo contacts match the original app's). 10 end-to-end tests in Playwright for Java 1.63.0, one per workflow that passed the old vs new comparison. They need a freshly started app, run in a fixed order, and add about 200 MB to every Maven repository and to the Docker image, because every build compiles them. See section 3.4.
- **CI:** one GitHub Actions job builds, starts the app in production mode, smoke-checks it, runs the end-to-end tests and fails on any error in the server log (section 3.5).

3 Environment, build, run, test and data

You need JDK 21 and Maven 3.9; Node 22 with npm for the CI steps; Docker for the reproducible environment. No Vaadin licence key and no database are needed (section 3.6). Full details are in `README.md`, `env/README.md` and `ci/README.md` in the repository.

3.1 Production build and run

```
mvn package          # production war in target/; runs the unit tests
mvn jetty:run-war    # builds and runs it on http://localhost:8080/
```

The log line "Vaadin is running in production mode." confirms the mode. To deploy elsewhere, put `target/addressbook-2.0.war` on a Jakarta EE 11 / Servlet 6.1 container. The app is a demo and has no production deployment target.

3.2 Development mode

```
mvn jetty:run        # development mode, http://localhost:8080/
```

Development mode adds Vaadin's dev tools. Usage statistics are off (set in `pom.xml`). It compiles the tests but does not run them.

3.3 Docker environment

```
env/up.sh            # build the image, start a copy on port 8080
env/reset.sh         # back to the seed contacts (restart)
env/down.sh          # stop and remove the copy
```

`APP_ENV_PORT`, `APP_ENV_PROJECT` and `APP_ENV_CONTAINER` run several copies side by side, each with its own data; `APP_ENV_PRODUCTION_MODE=false` runs development mode. The port is bound to 127.0.0.1 only. See `env/README.md`.

3.4 Tests

```
mvn test             # unit tests
mvn jetty:run-war    # or env/up.sh: a freshly started app
# then, in a second terminal, the end-to-end tests:
mvn verify -Pe2e -Dapp.url=http://localhost:8080
```

The contacts are in memory and the tests change them, so restart the app before each end-to-end run; a second run against the same app fails. Playwright downloads Chromium on the first run; set `PLAYWRIGHT_BROWSERS_PATH` and `PLAYWRIGHT_SKIP_BROWSER_DOWNLOAD=1` to use browsers you already have. Failure screenshots go to `target/e2e-failures/`.

The workflow "Enter a birth date with the calendar and by typing" has no end-to-end test yet. It was left out so that the reviewed test set did not change after its review; adding it, with its calendar step checking the accepted position, is follow-up work (section 6).

3.5 CI

`.github/workflows/ci.yml` runs one job on `ubuntu-24.04` on every push to `main`, `mig-dev`, `stage/**`, `stabilise/**` and `fix/**` and on every pull request. It needs no secrets. In order: JDK 21, `mvn -B package`, Node 22 and Playwright with Chromium, `ci/start-app.sh` (production mode), the smoke check (the contact list must show rows, with no console or page errors), the production-mode check, the end-to-end tests, the server-log check (any `SEVERE` or `ERROR` line or stack trace fails the job), and stop. On failure only, logs and screenshots are kept for 3 days. A run takes a few minutes. `ci/README.md` lists the same steps to run locally.

Playwright is pinned twice: in `pom.xml` (the end-to-end tests) and in `ci/package-lock.json` (the smoke check and the browser install). The tests use the browser that CI installs, so the two versions must be changed together.

3.6 Database

None. The app has no database. Its 100 demo contacts are generated in memory at start from a fixed seed and are shared by every user until the app restarts. There are no migration scripts.

4 Known pre-existing issues

These issues were in the stabilised Vaadin 8 app that the new app was compared with, and were carried over unchanged, so that the new app behaves the same. Most were in your original app; the CI and environment script issues at the end came with the CI and Docker environment added while preparing the old app, before the move to Vaadin 25. These are the ones a user or developer is likely to notice. The full list (36 items, including build and tooling details) is available on request.

Users notice:

- All users share one contact list: one user's changes are visible to everyone until the app restarts. Another user's changes only appear after your own save, filter or reload. When two users edit the same contact, the last save wins without a warning.
- There is no Delete button; a contact can only be removed by restarting the app (which resets all data).
- There are no input checks: names can be empty and email and phone take any text, so an empty new contact saves as a blank row.
- When a birth date is not a valid date, Save does nothing and gives no message about the save (the new app does show "Date format not recognized" at the field; section 2.1).
- Cancel on a new contact shows "Cancelled" but leaves the form open with what was typed.
- The filter searches the whole contact record, not only the shown columns: it also matches the phone number, the birth date in English text and internal field names (typing `name` matches every contact).
- Enter saves the open contact even when the cursor is in the filter field.
- Birth dates show two-digit years, and future birth dates are accepted.

Developers notice:

- commons-beanutils 1.9.2 (CVE-2019-10086, CVE-2025-48734) and, through it, commons-collections 3.2.1 (CVE-2015-6420 / CVE-2015-7501). Kept at the agreed version; the app copies only its own beans with it and deserialises nothing.
- The contact service is a lazily created singleton that is not thread-safe at start-up, and it is not serialisable, so sessions cannot be persisted or clustered.
- CI and env scripts: GitHub Actions pinned to major tags instead of commit hashes; `ci/start-app.sh` accepts any server answering on its port; the docs do not list `curl`, `sha256sum`, `find` and `xargs` as prerequisites; env copies run without a memory limit.

5 Old and new side by side

The 11 user workflows recorded on the old app were run on the new app step by step and compared screenshot by screenshot. The walkthrough shipped next to this document shows each workflow with the old and new screenshots of every step and a note on each difference: docs/walkthrough/index.html (open it in a browser; it needs no server). It shows all 11 workflows, and each difference you accepted on 23 October 2026 is captioned as accepted.

Result: 11 of 11 workflows meet the acceptance bar: 8 step for step, 3 with accepted differences.

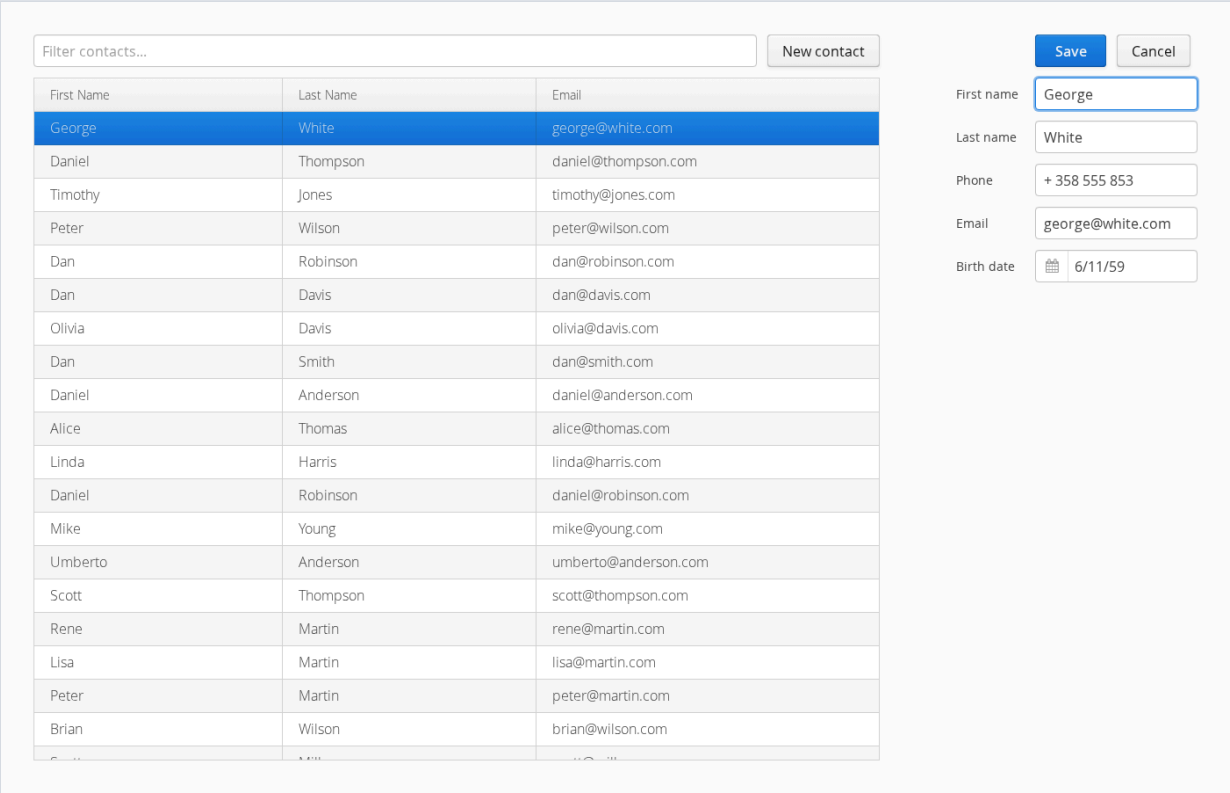


Figure 2. The old app (Vaadin 8.14.5, Valo) with the same contact open, 1400 x 900 window. Compare with figure 1.

Workflow	Result
Browse the contact list; filter it; open and close a contact; add, edit, cancel; save with missing values; see another user's changes	Same behaviour. About one more row is visible with Lumo (accepted with the theme)
Try to save an invalid birth date	Same result, with an accepted difference: the message shows under the field
Two users edit the same contact	Same result, with an accepted difference: the "Saved" message closes by itself after 3 s
Enter a birth date with the calendar and by typing	Same result, with an accepted difference: the calendar for a new contact opens on the last month shown, and the calendar has its new layout

6 Possible follow-up work

Improvements that are not needed for the migration and were left out on purpose. Each is small. They are input for a follow-up, not part of this delivery.

Look (the migration aimed at the same structure, not the same pixels; checked at 1400, 1024 and 1920 px wide unless noted):

- Lumo differs from Valo by design: grey-filled fields, borderless buttons, a taller grid header (56 px, old 38), pale row selection, no column lines or zebra rows, a pink fill for an invalid field, system font, white background, different sort and calendar icons (checked at 1400 px). A restoration step can bring the old look back where wanted.
- Save and Cancel sit 12 px lower, the grid starts 12 px lower, the gap between list and form is 16 px (old 12 px), form fields are 192 px wide (old 185 px).
- Notification text is dark on white (old blue). No tab icon: add a `favicon.ico` if wanted.

CI speed:

- Cache the Chromium install, or use a runner image that has it (one early run spent 9.5 minutes installing it; usually about 15 s).
- Run the end-to-end tests without rebuilding the war while the app is running.
- Let CI start the app through `env/up.sh`, so there is one way to start it instead of two.

Tests:

- Replace the fixed waits in the end-to-end tests and the wait on Vaadin's internal client object with waits on what the page shows.
- Give the demo-data unit test its own contact service, so test order cannot matter later.
- Add the end-to-end test for the birth-date workflow: its steps as on the old app, with the calendar step checking the accepted position. It is the only workflow without one.

Dependencies and code:

- Replace the bean copy by a small copy method so commons-beanutils and commons-collections, with their known vulnerabilities, can be removed.
- Store the birth date as a date without time (`LocalDate`) instead of converting in the form; this removes the time-zone and time-of-day handling.
- Run the war on a slim Jetty image instead of Maven and a JDK in the Docker image (smaller and faster to pull).

7 Decisions made

On 23 October 2026 you answered all 21 items, each as recommended. No decision is open.

7.1 Behaviour and look differences

#	Difference	Decision (23 October 2026)
1	Calendar for a new contact opens on the month last shown in the tab	Option A: accepted; the prepared fix is not taken, so the app relies on no internal part of Vaadin's date picker
2	A typed two-digit year in the boundary year is read by year, not by day	Accepted
3	The date order is always month/day/year	Accepted
4	Clicking a field's label puts the cursor in the field	Accepted
5	The new calendar popup layout	Accepted
6	"Saved" and "Cancelled" close by themselves after 3 s	Accepted
7	"Date format not recognized" shows under the field	Accepted
8	With the calendar open, one Enter picks the day and saves	Accepted
9	The browser's default tab icon	Accepted; adding an icon is follow-up work (section 6)

With these, the workflows "Enter a birth date", "Try to save an invalid birth date" and "Two users edit the same contact" pass with accepted differences.

7.2 Changes made by the migration

#	Change	Decision (23 October 2026)
10	Java 21 and Jetty 12.1 (Servlet 6.1, Jakarta EE 11)	Accepted
11	A built war always runs in production mode; the env switch takes only `true` or `false`	Accepted
12	New server log format and three harmless Jetty warnings at start	Accepted
13	Renamed and larger Docker image (`addressbook-new`)	Accepted
14	The stricter CI	Accepted
15	The Vaadin frontend build step and Java 21 compilation	Accepted
16	Ten end-to-end browser tests and their test library	Accepted
17	No usage statistics from development runs	Kept off
18	The Vaadin 25 dependency set (Apache 2.0 or MIT; the server-provided servlet API EPL 2.0 or GPLv2 with the Classpath Exception; JUnit EPL 2.0, tests only)	Accepted
19	Vaadin's development tools on the development classpath only	Accepted

#	Change	Decision (23 October 2026)
20	The restored layout	Accepted

7.3 Where the migrated code ends up

#	Question	Decision (23 October 2026)
21	Where the migrated app ends up	Option A: the Vaadin 25 app stays on `mig-dev`, beside the old app on `main`. Both stay runnable and can be compared side by side; nothing is merged into `main`

7.4 Acceptance

With these answers, the remaining steps are a walkthrough of this document with you and your acceptance of the migration.